Density altitude and lift

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Three steps

1 · Pressure altitude
PA = elevation + (29.92 − altimeter) × 1,000

2 · Standard temperature
ISA = 15 °C − 2 °C per 1,000 ft of PA

3 · Density altitude
DA ≈ PA + 120 × (OAT − ISA)

Worked example

Field elevation 1,400 ft, altimeter 29.82, OAT 30 °C.

Pressure altitude	1,400 + (29.92 − 29.82) × 1,000 = 1,400 + 100 = **1,500 ft**
Standard temperature	15 − 2 × 1.5 = **12 °C**
Density altitude	1,500 + 120 × (30 − 12) = 1,500 + 2,160 = **3,660 ft**

3,660 ft
The balloon performs as if the field were 2,260 ft higher.

What it does to lift

- Buoyancy equals the weight of the air the envelope displaces. Heating the air inside makes it less dense, so the balloon rises when it weighs less than the air it pushes aside.
- High elevation, low pressure, high temperature and high humidity all mean thinner air and less lift at the same envelope temperature: more heat for the same load, more fuel, less margin under the envelope's limit.
- Rough rule: 1,000 cubic feet of heated air lifts about 20 lb.
- Burner performance drops about 4% per 1,000 ft.
- Maximum gross weight is in your flight manual; its performance chart reads lift from ambient temperature, envelope temperature and pressure altitude.

Worksheet

Step	Flight 1	Flight 2
Field elevation (ft)		
Altimeter setting (inHg)		
(29.92 − altimeter) × 1,000		
Pressure altitude		
ISA = 15 − 2 × (PA ÷ 1,000)		
OAT (°C)		
(OAT − ISA) × 120		
Density altitude		

The 120 ft per °C step is a common rule of thumb, an approximation. For PA 10,000 ft at 20 °C it gives 13,000 ft; the Aviation Weather Handbook's computed answer is 12,700 ft. A flight computer or the app's chart is closer.

Sources: FAA-H-8083-25C Ch. 4 (standard atmosphere, pressure and density altitude), FAA-H-8083-11B Ch. 2 (buoyancy), FAA-H-8083-11B Ch. 3 (performance planning), FAA-H-8083-28B Ch. 8 (density altitude example). Original card, not an FAA document. Study aid, not for navigation.